

Portfolio Backtest Evidence: 1997-2026

Net USD evidence for every current portfolio output, regional index benchmarks, and an independent point-in-time index challenger.

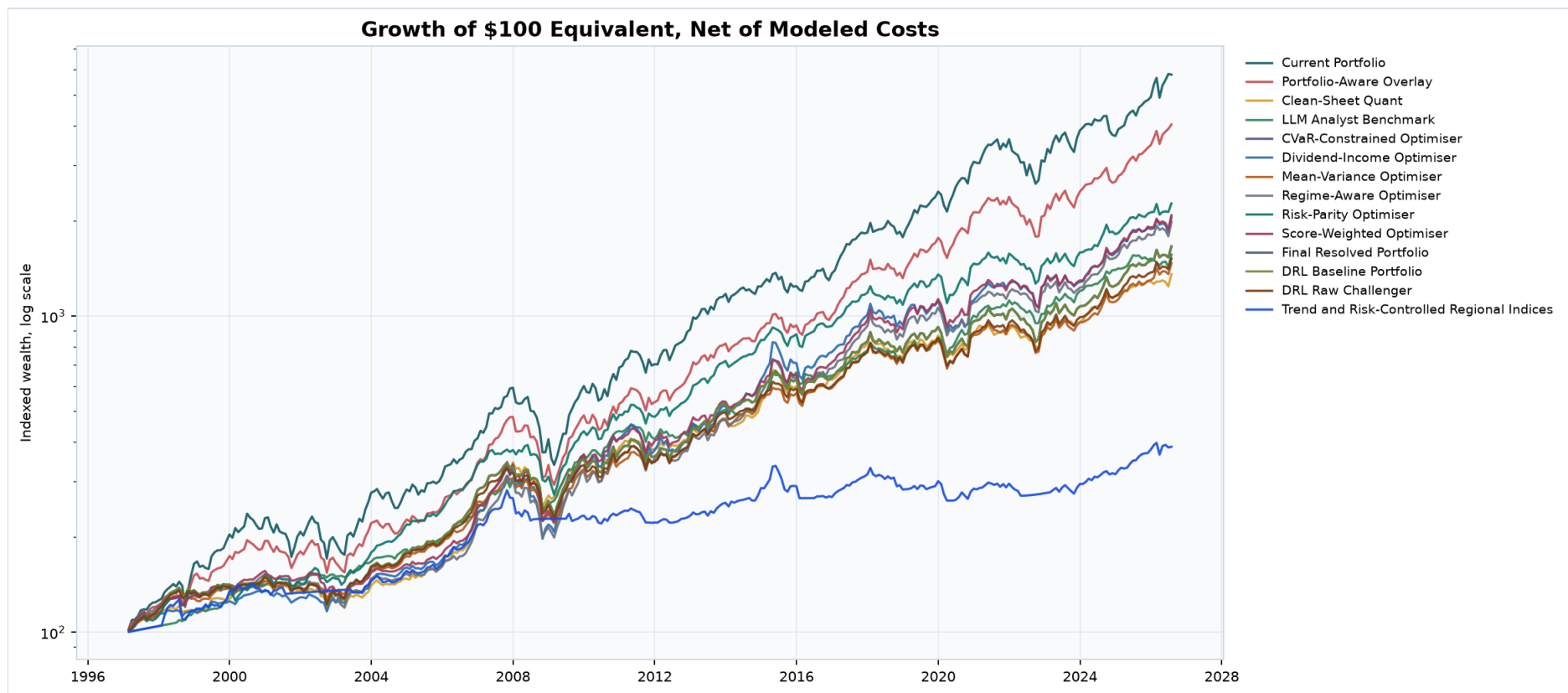
The long-history portfolio results are retrospective holdings replays, not point-in-time stock-selection backtests. Use them for exposure and path diagnostics, not as evidence that the current model could have selected these names in 1997.

13 portfolio outputs tested	29.6 requested historical years	\$80,540 current-derived starting NAV	2013-09-30 common 80% investable start	25 separate reconstructed PIT months
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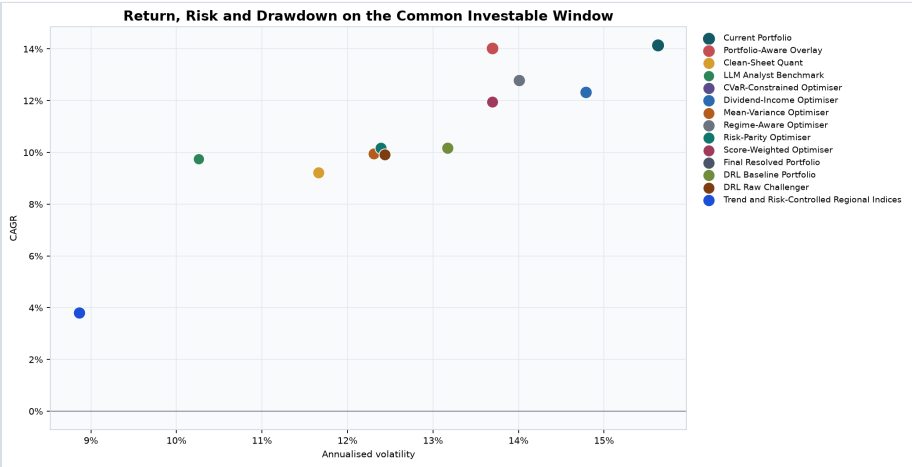
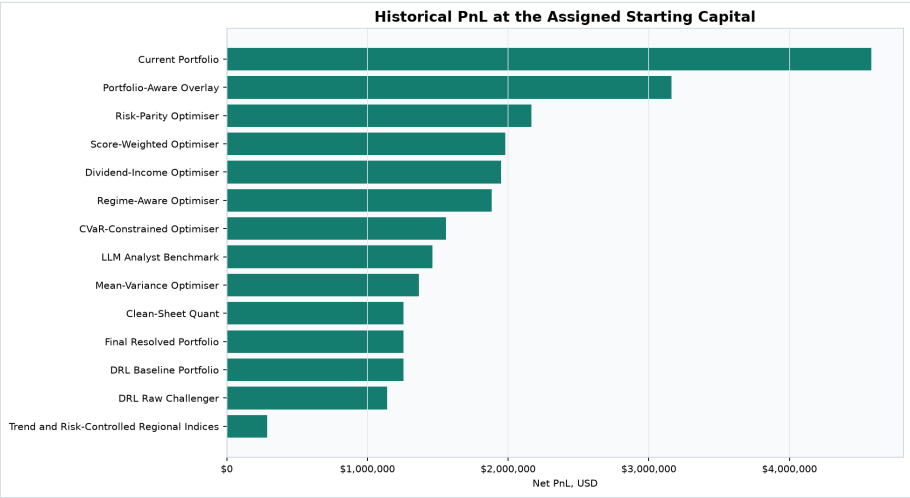
Executive Comparison

Portfolio	Start	cagr	Volatility	sharpe	Max drawdown	Ending value	PnL
LLM Analyst Benchmark	\$100,000	9.77%	9.10%	0.83	-20.90%	\$1,563,108	\$1,463,108
Portfolio-Aware Overlay	\$80,540	13.34%	14.34%	0.80	-39.00%	\$3,242,312	\$3,161,772
Risk-Parity Optimiser	\$100,000	11.16%	11.56%	0.79	-30.39%	\$2,266,941	\$2,166,941
Current Portfolio	\$80,540	14.75%	17.10%	0.77	-42.94%	\$4,661,466	\$4,580,926
Score-Weighted Optimiser	\$100,000	10.84%	12.72%	0.71	-35.20%	\$2,081,039	\$1,981,039
Mean-Variance Optimiser	\$100,000	9.53%	11.03%	0.69	-29.24%	\$1,465,240	\$1,365,240
Clean-Sheet Quant	\$100,000	9.24%	10.72%	0.68	-27.73%	\$1,355,449	\$1,255,449
Regime-Aware Optimiser	\$100,000	10.66%	13.08%	0.68	-36.43%	\$1,982,460	\$1,882,460

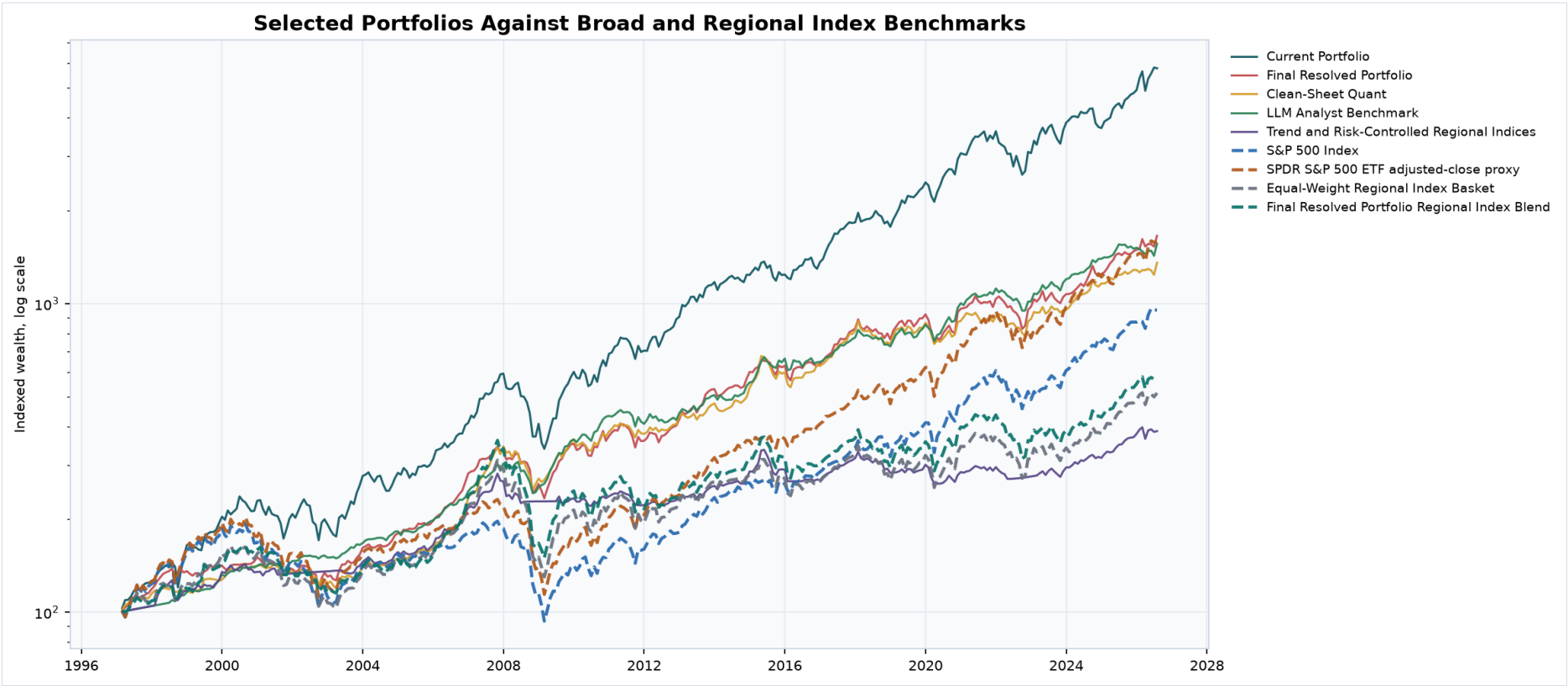
Portfolio	Start	cagr	Volatility	sharpe	Max drawdown	Ending value	PnL
DRL Raw Challenger	\$80,540	9.66%	11.47%	0.68	-30.55%	\$1,221,878	\$1,141,338
CVaR-Constrained Optimiser	\$100,000	9.99%	12.18%	0.67	-31.93%	\$1,657,883	\$1,557,883
DRL Baseline Portfolio	\$80,540	9.99%	12.18%	0.67	-31.93%	\$1,335,175	\$1,254,635
Final Resolved Portfolio	\$80,540	9.99%	12.18%	0.67	-31.93%	\$1,335,175	\$1,254,635
Dividend-Income Optimiser	\$100,000	10.78%	14.14%	0.64	-41.14%	\$2,049,676	\$1,949,676
Trend and Risk-Controlled Regional Indices	\$100,000	4.68%	8.64%	0.32	-22.34%	\$385,838	\$285,838



Indexed growth uses each portfolio's assigned capital only for PnL; this chart rebases every series to 100 for comparability.



Historical Risk

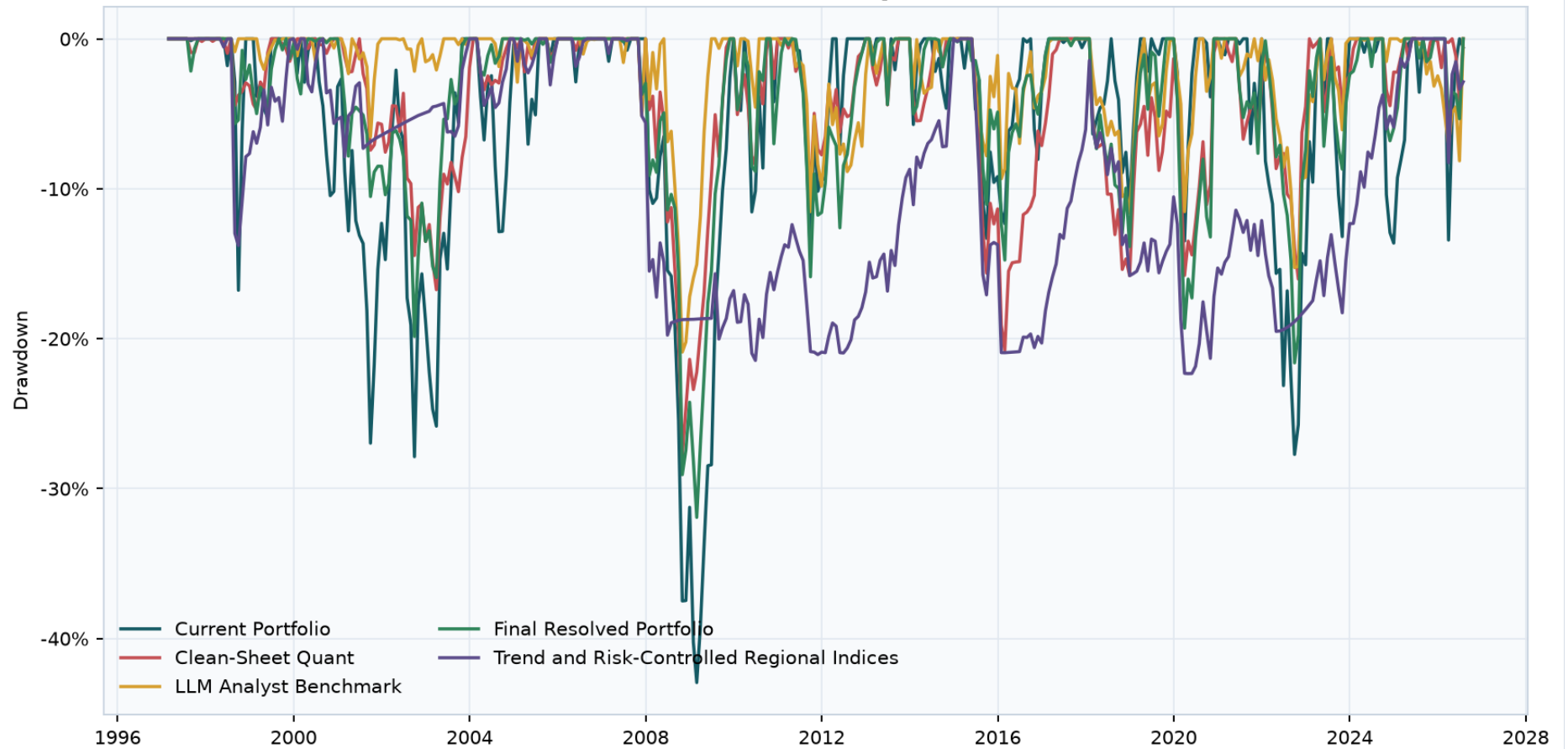


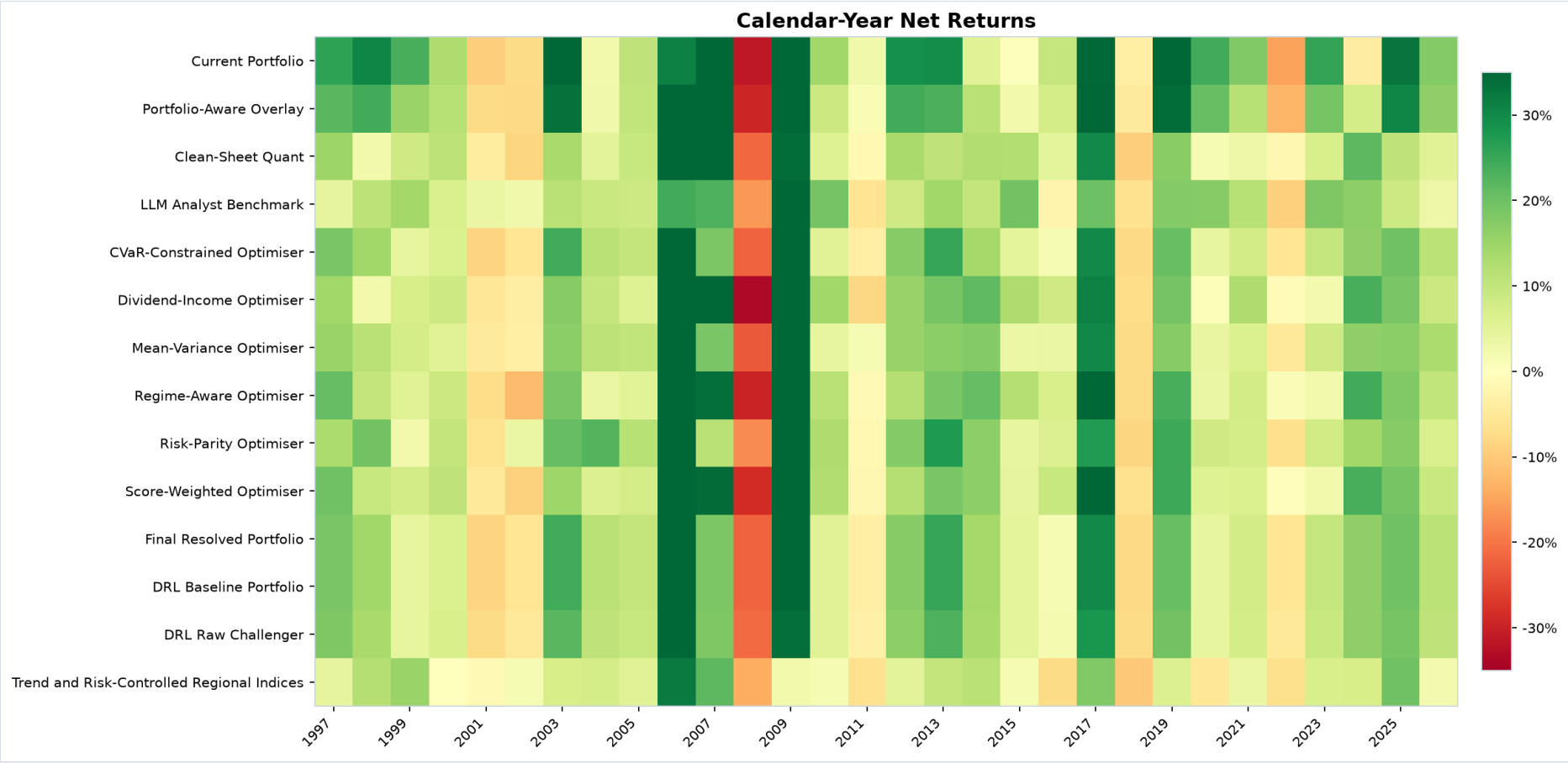
Dashed lines are broad or regional index benchmarks. Portfolio-specific relative statistics use each allocation's geographic index blend on the common investable window.

Primary Index Performance, Common Investable Window

Index benchmark	cagr	Volatility	sharpe	Max drawdown	Ending value on \$100k
SPDR S&P 500 ETF adjusted-close proxy	14.41%	14.44%	0.88	-23.93%	\$569,250
S&P 500 Index	12.52%	14.45%	0.77	-24.77%	\$458,656
Equal-Weight Regional Index Basket	6.20%	14.44%	0.36	-28.51%	\$217,529

Drawdown Comparison





Portfolio-Relative Regional Benchmark Results

Portfolio	Regional benchmark	annualised_alpha	beta	tracking_error	information_ratio	active_return_cumulative	Relative PnL
Portfolio-Aware Overlay	Portfolio-Aware Overlay Regional Index Blend	8.57%	0.80	0.067	1.07	159.47%	\$269,893
LLM Analyst Benchmark	LLM Analyst Benchmark Regional Index Blend	5.93%	0.88	0.059	0.91	97.94%	\$164,592
Current Portfolio	Current Portfolio Regional Index Blend	8.52%	0.84	0.082	0.90	160.34%	\$273,974

Portfolio	Regional benchmark	annualised_alpha	beta	tracking_error	information_ratio	active_return_cumulative	Relative PnL
Dividend-Income Optimiser	Dividend-Income Optimiser Regional Index Blend	7.23%	0.86	0.072	0.88	128.25%	\$252,289
Regime-Aware Optimiser	Regime-Aware Optimiser Regional Index Blend	7.03%	0.82	0.076	0.74	110.54%	\$248,363
Score-Weighted Optimiser	Score-Weighted Optimiser Regional Index Blend	6.79%	0.79	0.073	0.74	104.76%	\$219,865
Clean-Sheet Quant	Clean-Sheet Quant Regional Index Blend	5.33%	0.79	0.073	0.58	73.34%	\$132,219
DRL Raw Challenger	DRL Raw Challenger Regional Index Blend	4.83%	0.77	0.071	0.45	54.89%	\$96,796
DRL Baseline Portfolio	DRL Baseline Portfolio Regional Index Blend	4.93%	0.77	0.075	0.43	56.28%	\$101,361
Final Resolved Portfolio	Final Resolved Portfolio Regional Index Blend	4.93%	0.77	0.075	0.43	56.28%	\$101,361
CVaR-Constrained Optimiser	CVaR-Constrained Optimiser Regional Index Blend	4.92%	0.77	0.075	0.43	56.26%	\$125,815
Mean-Variance Optimiser	Mean-Variance Optimiser Regional Index Blend	5.10%	0.72	0.075	0.42	55.39%	\$121,354
Risk-Parity Optimiser	Risk-Parity Optimiser Regional Index Blend	4.09%	0.75	0.069	0.28	32.99%	\$86,582

The Three Backtests

1. Historical / walk-forward

Monthly execution with one-period signal lag for the index challenger.
Current portfolio outputs are shown only as retrospective holdings replays
and are explicitly flagged for selection look-ahead and survivorship bias.

2. Resampling

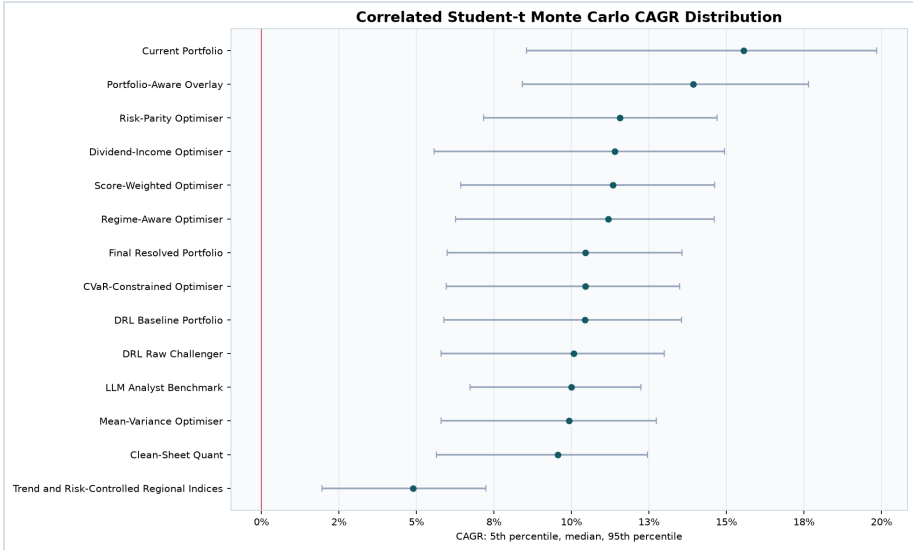
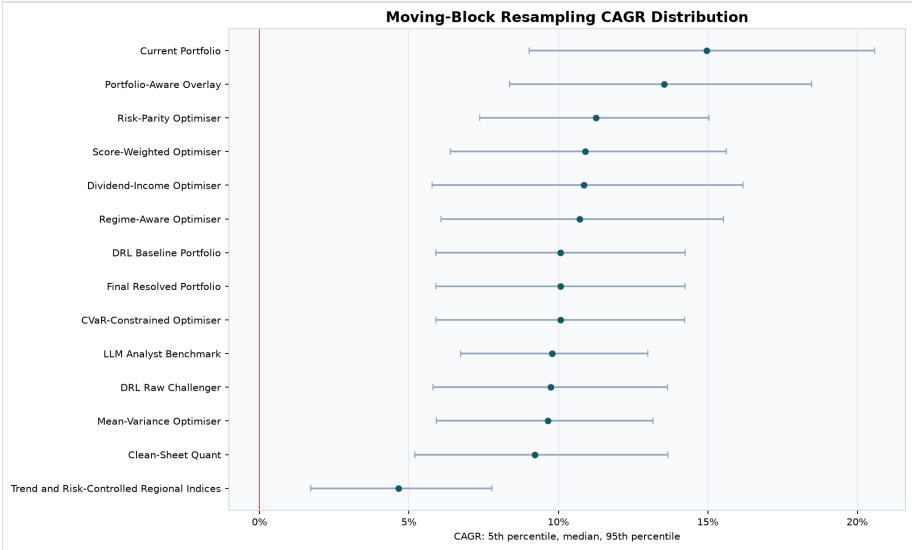
Circular 12-month moving blocks preserve serial dependence and cross-strategy alignment. The intervals show how path order changes CAGR, Sharpe, drawdown, and benchmark outperformance.

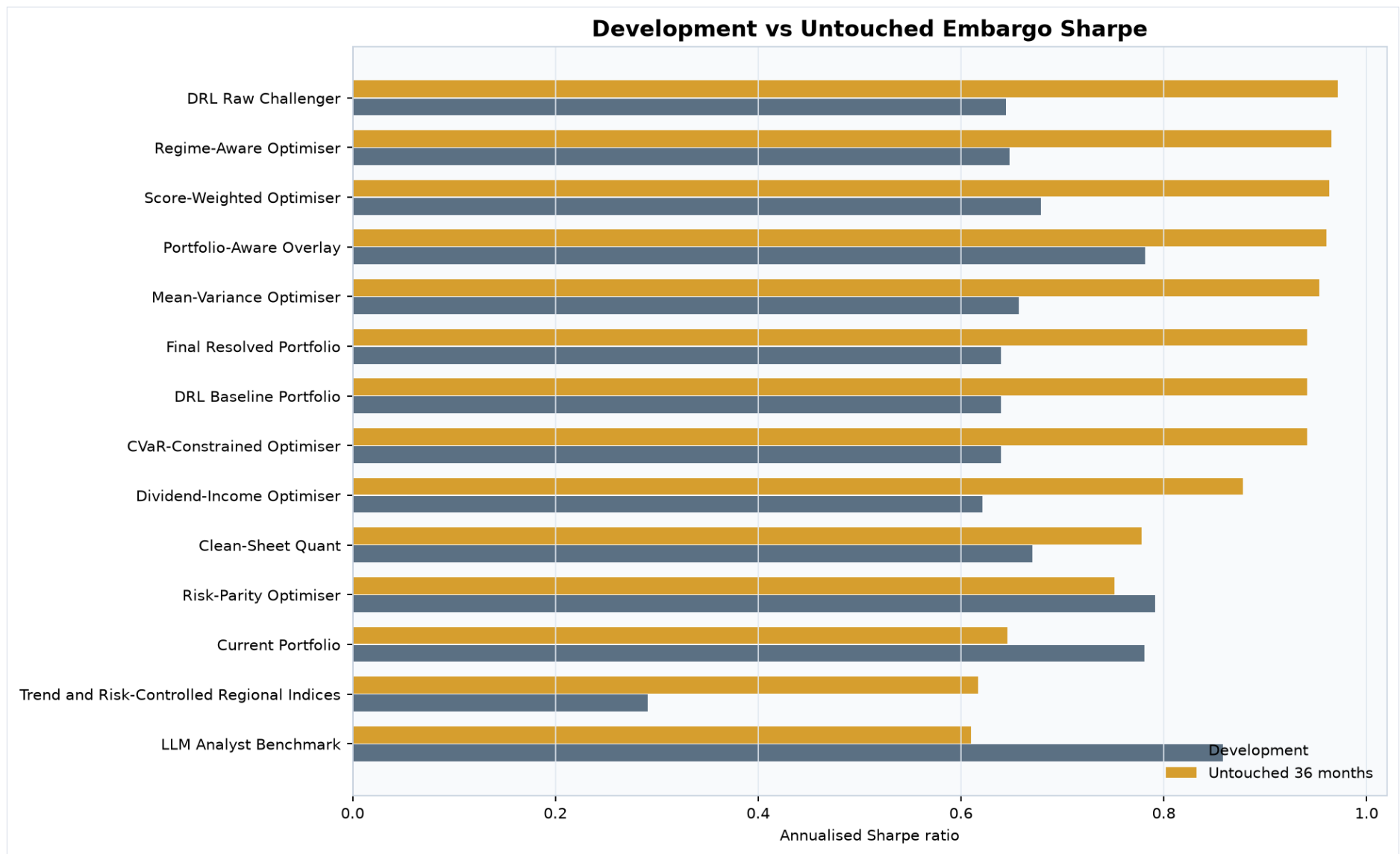
3. Monte Carlo

A correlated Student-t AR(1) process with EWMA conditional variance preserves cross-portfolio dependence, autocorrelation, fat tails, and volatility clustering without fitting a return target.

Statistical integrity

PSR, MinTRL, Sidak family-wise control, and Deflated Sharpe Ratios account for sample length, non-normal returns, and the number and correlation of portfolio trials.





Portfolio	annualised_sharpe	PSR	MinTRL months	sidak_significant	DSR	cluster_adjusted_deflated_sharpe_ratio
llm_benchmark	0.83	100.00%	50.998	True	99.95%	99.99%

Portfolio	annualised_sharpe	PSR	MinTRL months	sidak_significant	DSR	cluster_adjusted_deflated_sharpe_ratio
portfolio_aware_overlay	0.80	100.00%	57.712	True	99.88%	99.98%
optimised_risk_parity	0.79	100.00%	59.550	True	99.85%	99.98%
current_portfolio	0.77	100.00%	61.213	True	99.82%	99.97%
optimised_score_weighted	0.71	99.99%	69.679	True	99.57%	99.93%
clean_sheet	0.68	99.99%	71.291	True	99.47%	99.91%
optimised_mean_variance	0.69	99.98%	75.153	True	99.38%	99.89%
optimised_regime_aware	0.68	99.98%	74.444	True	99.37%	99.89%
drl_challenger_raw	0.68	99.98%	78.168	True	99.26%	99.86%
optimised_cvar_constrained	0.67	99.97%	79.991	True	99.17%	99.84%
drl_baseline	0.67	99.97%	79.994	True	99.17%	99.84%
final_portfolio	0.67	99.97%	79.994	True	99.17%	99.84%
optimised_dividend_income	0.64	99.97%	81.220	True	99.02%	99.82%
trend_risk_controlled_indices	0.32	95.38%	338.108	False	72.79%	87.59%

Data Quality and Causality

Execution and Liquidity

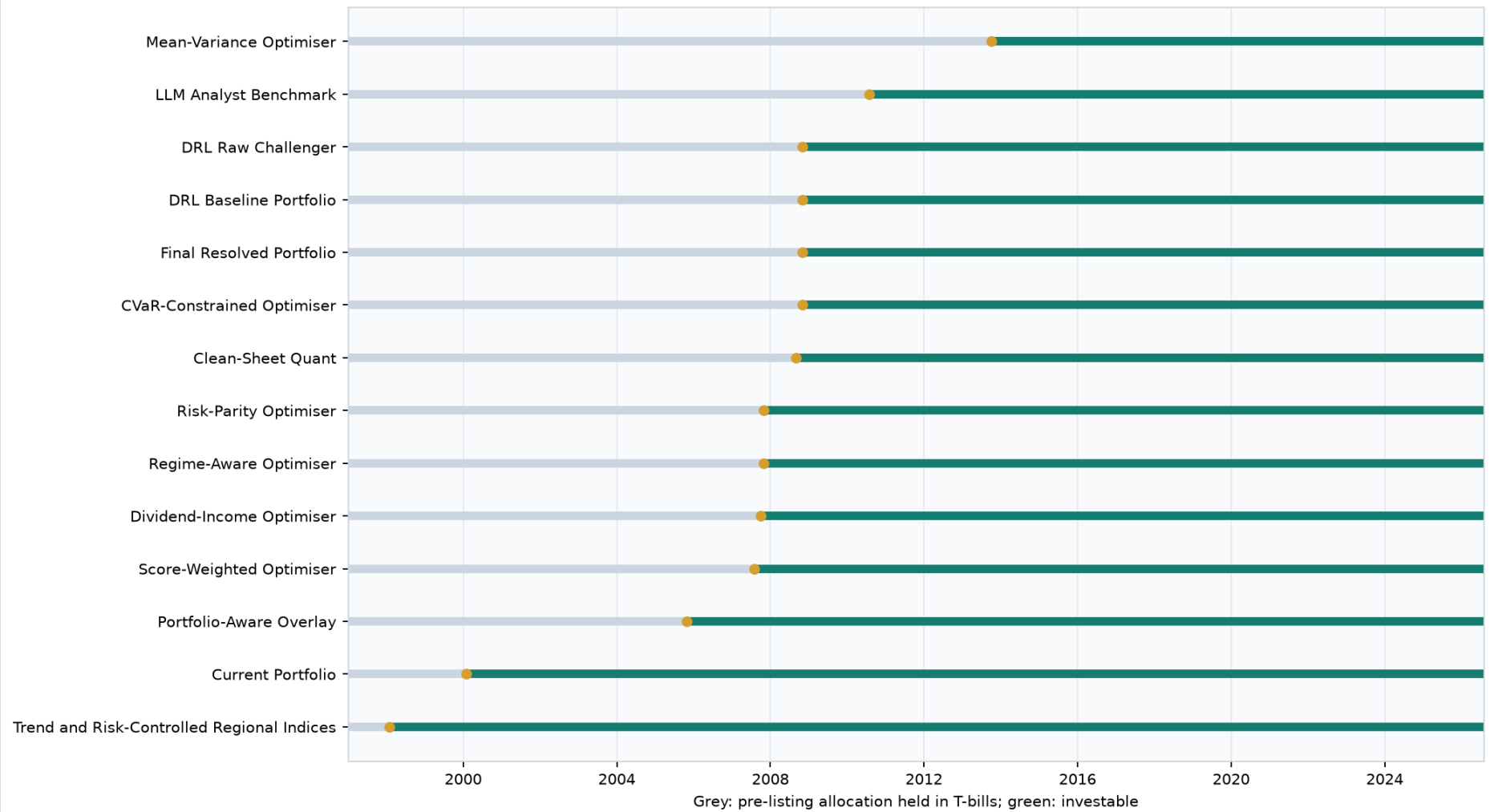
Portfolio	Total modeled cost	Annual turnover	Max ADV participation	Constrained trade events	Max unfilled target
current_portfolio	\$16,843	29.93%	1.31%	0	0.00%
trend_risk_controlled_indices	\$12,720	126.60%		0	0.00%
portfolio_aware_overlay	\$12,091	29.71%	0.55%	0	0.00%

Portfolio	Total modeled cost	Annual turnover	Max ADV participation	Constrained trade events	Max unfilled target
optimised_dividend_income	\$11,447	30.88%	5.00%	10	0.53%
optimised_risk_parity	\$11,163	27.98%	5.00%	1	0.82%
optimised_score_weighted	\$10,431	28.98%	5.00%	10	0.47%
optimised_regime_aware	\$9,883	29.43%	5.00%	10	0.46%
optimised_cvar_constrained	\$8,972	29.32%	5.00%	3	0.91%
llm_benchmark	\$8,503	24.14%	5.00%	62	4.43%
optimised_mean_variance	\$7,546	27.13%	5.00%	5	4.42%
drl_baseline	\$7,127	29.34%	5.00%	1	0.70%
final_portfolio	\$7,127	29.34%	5.00%	1	0.70%
drl_challenger_raw	\$6,267	28.40%	5.00%	18	0.53%
clean_sheet	\$6,083	25.90%	5.00%	11	4.05%

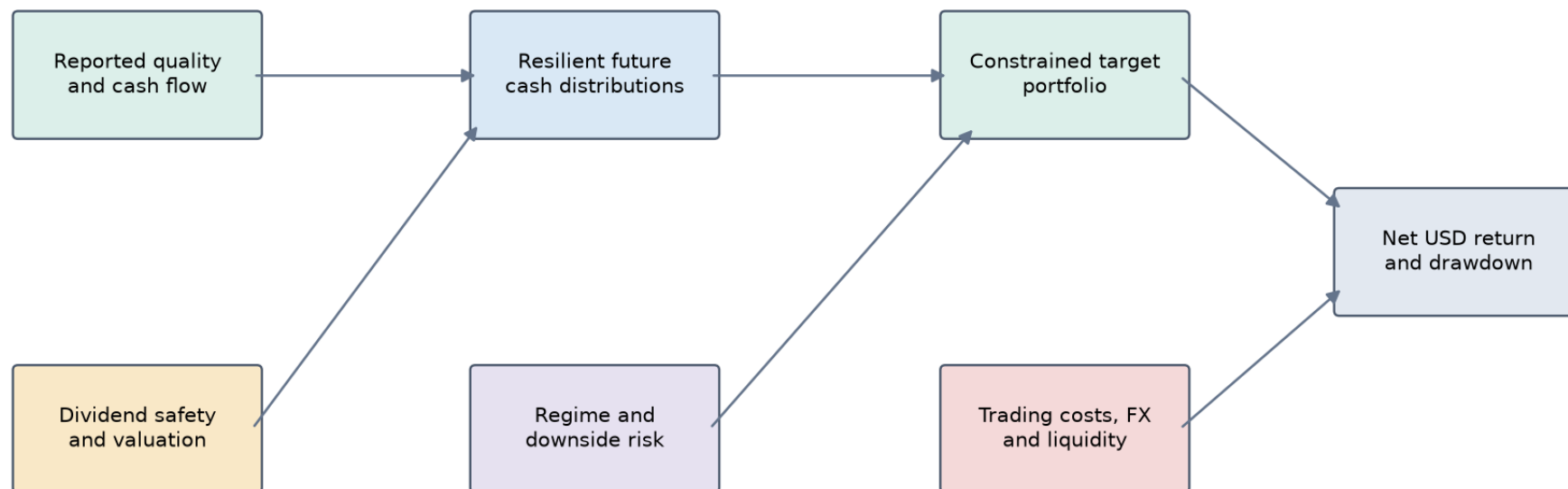
Adjusted-Price Repairs

symbol	Event date	Auditable repair	Raw return	Post-repair return	Scale factor
600018.SS	2006-10-26 00:00:00	persistent_level_shift_historical_rescale	-76.85%	0.00%	0.232
GEBN.SW	1999-06-22 00:00:00	persistent_level_shift_historical_rescale	-58.96%	0.00%	0.410
SHEL.L	1997-07-01 00:00:00	persistent_level_shift_historical_rescale	-71.74%	0.00%	0.283
TTE.PA	2005-12-26 00:00:00	isolated_spike_log_interpolation	300.00%	-0.16%	0.250
ULVR.L	1997-08-26 00:00:00	persistent_level_shift_historical_rescale	322.53%	0.00%	4.225
ULVR.L	1997-10-13 00:00:00	persistent_level_shift_historical_rescale	83.68%	0.00%	1.837

When at Least 80% of Each Intended Asset Allocation Was Tradable



Causal Theory Tested by the Model and Degraded by Execution Frictions



category	severity	applies_to	status	treatment
selection look-ahead	CRITICAL	all current portfolio-output replays	ACKNOWLEDGED	Labelled retrospective holdings replay; excluded from claims of 1997 model selection skill.
survivorship bias	CRITICAL	current selected securities	ACKNOWLEDGED	Pre-listing capital stays in T-bills; current holdings are never presented as a historical universe.
point-in-time model evidence	HIGH	full stock-selection strategy	SEPARATED	The 25-month reconstructed decision history is reported separately and never spliced into the long replay.

category	severity	applies_to	status	treatment
corporate actions	MEDIUM	holding histories	MODELED	Yahoo adjusted closes include splits and distributions where provider history supports them.
currency conversion	MEDIUM	non-USD holdings and indices	MODELED	Historical FRED FX converts local adjusted closes to USD; ECU bridges pre-euro observations.
benchmark dividend consistency	MEDIUM	regional price indices	WARNING	DAX is a performance index; other regional index series may omit dividends. SPY is included as a total-return proxy.
transaction costs and liquidity	MEDIUM	portfolio outputs and index challenger	MODELED	Commission, spread, slippage, square-root impact, missing-ADV penalty, and 5% ADV caps are reported.
short-sale constraints	LOW	all tested portfolios	NOT_APPLICABLE	Every tested allocation is long-only; no borrow availability or fee assumption is required.
multiple testing	HIGH	portfolio comparison	CONTROLLED	Sidak FWER and Deflated Sharpe Ratios use both actual and correlation-clustered trial counts.
non-stationarity	HIGH	all historical and simulated evidence	RESIDUAL_RISK	Embargo, subperiod plots, block resampling, fat tails, and EWMA volatility expose but cannot remove structural breaks.

Point-in-Time Model Evidence

The repository's reconstructed model walk-forward is not spliced into the 1997 holdings replay. It remains a separate, shorter evidence set because it uses dated model decisions and realised outcomes.

strategy	observations	annualised_return	annualised_volatility	sharpe	maximum_drawdown	annualised_cost_drag	evidence_mode
wolf_cvar	25	18.93%	10.84%	1.75	-5.85%	0.025	reconstructed_pit_proxy
equal_weight_eligible	25	24.99%	11.29%	2.21	-5.17%	0.006	reconstructed_pit_proxy
cap_weight_eligible	25	17.62%	8.88%	1.98	-3.99%	0.006	reconstructed_pit_proxy

Research evidence only. The paper used as the methodological specification is *The Three Types of Backtests* (Joubert, Sestovic, Barziy, Distaso, and Lopez de Prado, 29 July 2024). Raw Yahoo Finance and FRED histories remain outside Git; compact outputs, manifests, and checksums are included.